Diffusion of Oxygen Through Activated Sludge Flocs: Experimental Measurement, Modeling, and Implications for Simultaneous Nitrification and Denitrification

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ABSTRACT: Diffusion of dissolved oxygen through activated sludge flocs was studied, as it represents a potential mechanism for simultaneous nitrification and denitrification in activated sludge systems. Dissolved oxygen profiles through six floc particles collected at different times from a full-scale activated sludge plant demonstrated that that the dissolved oxygen concentration declines through all floc particles. For larger floc particles (2-mm diameter and greater), the dissolved oxygen concentration reached near-zero values at depths depending on process operating conditions. A mathematical model based on diffusion of dissolved oxygen, organic substrate (methanol), ammonia, nitrite, and nitrate through a spherical floc and consumption of dissolved oxygen by heterotrophs and autotrophs accurately predicted the dissolved oxygen profile and required adjustment of only one model parameter—the concentration of heterotrophs. A different dissolved oxygen decline pattern was exhibited for the smaller floc particles characterized, with the dissolved oxygen reaching a non-zero plateau toward the center of the floc. This pattern was not reproduced with the mathematical model developed and suggests that additional mechanisms are responsible for the transport of dissolved oxygen into the center of these flocs. Implications of these results regarding the occurrence of simultaneous nitrification and denitrification include consideration of the factors that affect floc size and distribution (simultaneous nitrification and denitrification is maximized with larger floc particles), coupling of the International Water Association (London) activated models to predict activated sludge composition with diffusion models to consider intrafloc effects, and the effects of substrate diffusion on the apparent half-saturation constant for various substrates in activated sludge systems. *Water Environ. Res.,* **79,** 375 (2007).

KEYWORDS: activated sludge, biological nutrient removal, nitrification, denitrification, simultaneous, diffusion, flocs.

doi:10.2175/106143006X111835

Introduction

Simultaneous nitrification and denitrification is a well-known phenomenon in biological nutrient removal activated sludge systems. As described in literature (Daigger and Littleton, 2000; Littleton, 2001, 2002; Littleton, Daigger, and Strom, 2003; Littleton, Daigger, Strom, and Cowan, 2003), several mechanisms can be responsible for its occurrence, but the two most likely are the following:

(1) Bioreactor macroenvironment. Anoxic and/or anaerobic zones may develop within the bioreactor, as a result of the mixing pattern caused, for example, by the oxygen transfer device.
(2) Floc microenvironment. Anoxic and/or anaerobic zones may develop inside the activated sludge flocs.

Diffusion of substrates through activated sludge flocs is a well-established phenomenon (Atkinson and Davies, 1974; Bakti, 1992; Benefield and Molz, 1984; Kossen, 1979; Matson and Characklis, 1976). This phenomenon is accounted for in the popular International Water Association (IWA) (London) activated sludge models (ASMs), as a result of the half-saturation constants in the Monod functions used in the microorganism growth rate expressions for heterotrophs for dissolved oxygen and nitrate (O'Shaugnessey et al., 1998). With proper choice of these model parameters, the IWA models can be tuned to account for simultaneous nitrification and denitrification occurring in full-scale activated sludge systems (Cinar et al., 1998; Daigger and Littleton, 2000). However, it is desirable to better characterize this phenomenon so that predictive models can be developed. This objective provided the motivation for this research.

In this research, a method for measuring dissolved oxygen profiles through activated sludge flocs was developed and used to test a mathematical model for dissolved oxygen diffusion and use through activated sludge flocs. Some implications of these results relative to the modeling of simultaneous nitrification and denitrification in activated sludge systems are subsequently addressed.

Materials and Methods

Development of Mathematical Model. A mathematical model for the diffusion and metabolism of dissolved oxygen, methanol, ammonia, nitrite, and nitrate through an activated sludge floc was developed in spherical coordinates. The reactions incorporated to the model include the following:

(1) Aerobic growth of heterotrophs on dissolved organic matter (methanol),
(2) Aerobic growth of ammonia-oxidizing autotrophs,
(3) Aerobic growth of nitrite-oxidizing autotrophs,
(4) Anoxic growth of heterotrophs using nitrate as a terminal electron acceptor,
(5) Anoxic growth of heterotrophs using nitrite as a terminal electron acceptor,

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Table 1—Biological process model in matrix format. Model and process reactions shown as exes for matrix. Model stoichiometry ($C_{i,j}$) listed in matrix. Kinetic expressions also shown.

		Constituents			
		Dissolved oxygen	Methanol	Ammonia	Nitrite
Process reactions		S_1	S_2	S_3	S_4
Aerobic growth of heterotrophs	e^- donor Methanol e^- acceptor Oxygen	$(1 - Y_H)$	1	i	0
Aerobic growth of ammonia-oxidizers	e^- donor Ammonium e^- acceptor Oxygen	$(3.43 - Y_A)$	0	$4.57 - i$	-1.145
Aerobic growth of nitrite oxiders	e^- donor NO_2^- e^- acceptor Oxygen	$(1.14 - Y_N)$	0	$3.98Y3 - i$	1.18
Aerobic decay of heterotrophs	e^- donor Biomass e^- acceptor Oxygen	$1 - Y_H$	0	0	0
Anoxic growth of heterotrophs with nitrite	e^- donor Methanol e^- acceptor Nitrite	0	1	i	$-(Y_{HNI}-1)_ 1.71$
Anoxic decay of heterotrophs with nitrite	e^- donor Biomass e^- acceptor Nitrite	0	0	0	$1 - Y_{HNI}$
Anoxic growth of heterotrophs with nitrate	e^- donor Methanol e^- acceptor Nitrate	0	1	i	0
Anoxic decay of heterotrophs with nitrate	e^- donor Biomass e^- acceptor Nitrate	0	0	0	0

(6) Aerobic decay of heterotrophs,
(7) Anoxic decay of heterotrophs using nitrate as a terminal electron acceptor, and
(8) Anoxic decay of heterotrophs using nitrite as a terminal electron acceptor.

Ammonia- and nitrite-oxidizer decay, under both aerobic and anoxic conditions, was omitted to simplify the model. Table 1 summarizes the kinetics and stoichiometry for the model in matrix format, while Table 2 presents values of the kinetic, stroichiometric, and diffusion parameters. Model parameters were largely taken from Bakti and Dick (1992), as described in Steller (2003). Model stoichiometry was developed based on microbial thermodynamics (Steller, 2003). Floc particles are assumed to be spherical, and boundary layer effects are considered to be negligible.

The model was implemented using the Scientist software program (MicroMath, Salt Lake City, Utah) as a system of simultaneously solved first-order ordinary differential equation according to the following equation:

$$D_{fj}\left(\frac{d^2S_j}{dr^2}+\frac{2dS_j}{rdr}\right) = \pm\sum_{i=1}^{n} C_{i,j}r_i \qquad (1)$$

Where

D_{fj} = diffusion coefficient for element j,
S_j = concentration of a diffusing substrate,
r = floc radius,
$C_{i,j}$ = stoichiometric coefficient for reaction i with component j, and
r_i = rate for reaction i (Steller, 2003).

The error-controlled Runge-Kutta numerical method was used to solve the model. Only the aerobic (dissolved oxygen available) portion of the model was tested experimentally, as data on only the dissolved oxygen profile through activated sludge flocs were obtained. The model was developed after Bakti (Bakti, 1988; Bakti and Dick, 1992). Further details are available in Steller (2003).

A sensitivity analysis was conducted, which classified model parameters into four categories (high, moderate, low, and insensitive) using the following sensitivity ratio:

$$\text{Sensitivity Ratio} = \frac{\%\ \text{Change of Constituent}}{\%\ \text{Change of Parameter}} \qquad (2)$$

Table 3 summarizes the results, which are discussed below.

Experimental Apparatus. *Dissolved Oxygen Microprobes and Micromanipulator.* Dissolved oxygen microprobes manufactured by Unisense (Unisence ApS, Science Park Aarhus, Denmark) were used to measure dissolved oxygen profiles through activated sludge particles, as illustrated in Figure 1. The microprobe is a Clark-type electrode with a guard electrode. Probes with tip diameters between 10 and 40 μm were used in this research. The 90% response time for the probe ranged from 0.1 to 20 seconds, and its detection limit was 0.003 mg/L, as reported by the manufacturer. The stirring sensitivity is less than 1% difference. Electrical interferences were significant and were overcome using a grounding wire from a grounded source directly into the solution being measured. Current from the electrode was measured using a fast-response (50 milliseconds for 90% response time) Model PA2000 picoammeter (Unisense) and was further recorded using Labview data acquisition software (National Instruments Corporation, Austin, Texas). The data were subsequently exported to Microsoft Excel for subsequent analysis (Microsoft Corporation, Bellevue, Washington). The microprobe was moved through the activated sludge floc using a micromanipulator (Unisense) capable of moving the probe by as little as 5-μm increments.

Reactor Design. A reactor was constructed to trap activated sludge flocs and to recirculate solution with defined properties past the floc to maintain a uniform environment. Figure 2 illustrates the reactor, which consisted of a bulk chamber (3.8-L volume, mixed to provide uniform bulk properties, i.e., dissolved oxygen) and a floc-capture chamber (127 mL), where the solution was continuously passed by a floc particle captured on a monofilament grid, as shown in Figure 3. Dissolved oxygen in the bulk chamber was measured

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Table 1—Extended.

Nitrate	Heterotrophs	Ammonia oxidizers	Nitrite oxidizers	Kinetic expressions
S_5	X_H	X_A	X_N	r_j
0	$-Y_H$	0	0	$k_1 X_H \left(\frac{S_1}{K_{s1}+S_1}\right)\left(\frac{S_2}{K_{s4}+S_2}\right)$
0	0	$-Y_A$	0	$k_2 X_A \left(\frac{S_1}{K_{s2}+S_1}\right)\left(\frac{S_3}{K_{s5}+S_3}\right)$
−2.86	0	0	Y_N	$k_3 X_N \left(\frac{S_1}{K_{s3}+S_1}\right)\left(\frac{S_4}{K_{s6}+S_4}\right)$
0	$1 - Y_H$	0	0	$k_4 X_H \left(\frac{S_1}{K_{s1}+S_1}\right)$
0	$-Y_{HNI}$	0	0	$k_5 X_H \left(\frac{S_2}{K_{s4}+S_2}\right)\left(\frac{K_{S1}}{K_{s1}+S_1}\right)$
0	$-Y_{HNI}$	0	0	$k_6 X_H \left(\frac{K_{s1}}{K_{s1}+S_1}\right)$
$-(Y_{HNA}-1)_ 2.86$	$-Y_{HNA}$	0	0	$k_7 X_H \left(\frac{S_2}{K_{s4}+S_2}\right)\left(\frac{K_{S1}}{K_{s1}+S_1}\right)$
$1- Y_{HNA}$	$1 - Y_{HNA}$	0	0	$k_8 X_H \left(\frac{K_{s1}}{K_{s1}+S_1}\right)$

using two Corning model 317 probes (Corning, Inc., Corning, New York). Aeration was manipulated to achieve the desired bulk dissolved oxygen. A peristaltic pump was used to recirculate the bulk solution at a flowrate of 5.1 mL/min, providing a flow velocity of 0.8 cm/min in the floc chamber to maintain constant conditions for the trapped floc particle. The hydraulic residence time in the floc capture chamber was 25 minutes. Recirculation was maintained for 50 minutes before dissolved oxygen profiles were measured. This time was based on the time required for diffusion of oxygen through the largest floc particles studied (i.e., 4 mm) using the smallest diffusion coefficient reported in the literature (Steller, 2003), with an additional safety factor of three. The actual size of the floc particles that were profiled ranged from 0.25 to 3 mm. Floc particles were captured with a 1-mL pipette from a seed sample collected from the Southeast Rolla (Missouri) Wastewater Treatment Plant and gently placed onto the monofilament grid. A 1-mL sample was transferred. The Southeast Rolla Wastewater Treatment Plant is an extended aeration activated sludge plant, with no primary clarification, that treats predominantly municipal wastewater. After steady-state conditions were achieved, the microprobe was moved through the captured floc at increments of either 0.005 or 0.01 mm in the downward direction to collect profile data. The reactor temperature was maintained at 22°C (+/−2°C). Figure 4 provides a photograph of the entire experimental apparatus.

Feed Solution. Table 4 summarizes the components of the synthetic solution used. The concentrations of methanol and ammonia-nitrogen were varied, depending on experimental objectives. Feed solution was prepared daily using prepared stock solutions and diluting with tap water to achieve the desired chemical oxygen demand (COD) and nitrogen concentrations. The concentrations of constituents were sufficiently high to be in excess of the biosynthetic requirements for all COD concentrations used. Biodegradation and volatilization of methanol from the feed solution were confirmed to be negligible by COD measurements of freshly prepared feed solution and solutions after 1 day, which confirmed that methanol loss was less than 4%.

Analytical Methods. The COD, ammonia, nitrate, and nitrite were measured with a Hach DR/2010 (Hach Company, Loveland, Colorado) using methods 8000, 10031, 10020, and 8507, respectively. Total and volatile suspended solids were measured using *Standard Methods* 2540D and 2540E, respectively (APHA et al., 1995).

Results

Initial intrafloc dissolved oxygen profiles developed in floc particles from the Southeast Rolla Wastewater Treatment Plant demonstrated that the dissolved oxygen profile through the floc depended on external dissolved oxygen concentration, among other factors, and that the dissolved oxygen inside of the floc would approach zero when the bulk (or external) dissolved oxygen concentration was sufficiently low (data not shown). The results also demonstrated that insertion of the probe created a channel that resulted in a different dissolved oxygen profile when the probe was withdrawn than when it was inserted. The dissolved oxygen profile when the probe was subsequently reinserted was also different than for the first insertion. Thus, only profiles from the first insertion are reported here. For more details, see Steller (2003).

Collection of reliable dissolved oxygen profiles proved to be quite tedious. Consequently, six profiles from four activated sludge samples (referred to as samples A through D) collected at distinctly different times were measured. For activated sludge samples A and B, dissolved oxygen profiles for two different floc particles were obtained, while, for samples C and D, only one floc particle from each activated sludge sample was profiled. For samples A and B, the mathematical model was calibrated to one dissolved oxygen profile, and the calibrated model was then used to predict the profile for the other floc particle (that is, one floc was used for model calibration, while the other was used for model verification). The sample A and B models were subsequently compared with the sample C and D results. Default model parameters were used for all but the bulk dissolved oxygen, methanol, and ammonia concentrations (meaning the concentrations in the bulk fluid, assumed to also

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Table 2—Definitions of model parameters and default values.

Parameter	Definition	Default values	Units	
R	Radius of floc particle	0.1	cm	L
i	Concentration of nitrogen in products from biomass	0.0826	g nitrogen/g COD in active biomass	M/M
D1	Diffusivity of dissolved oxygen in microbial floc	1.025	cm^2/d	L^2/T
D2	Diffusivity of methanol in microbial floc	0.666	cm^2/d	L^2/T
D3	Diffusivity of ammonium in microbial floc	0.769	cm^2/d	L^2/T
D4	Diffusivity of nitrite in microbial floc	0.7175	cm^2/d	L^2/T
D5	Diffusivity of nitrate in microbial floc	0.7175	cm^2/d	L^2/T
D6	Diffusivity of heterotrophs in microbial floc	0.1	cm^2/d	L^2/T
D7	Diffusivity of ammonia oxidizers in microbial floc	0.1	cm^2/d	L^2/T
D8	Diffusivity of nitrite oxidizers in microbial floc	0.1	cm^2/d	L^2/T
SC1	Concentration of oxygen at floc surface	—	mg/L	M/L^3
SC2	Concentration of methanol at floc surface	—	mg/L	M/L^3
SC3	Concentration of ammonium at floc surface	—	mg/L	M/L^3
SC4	Concentration of nitrite at floc surface	—	mg/L	M/L^3
SC5	Concentration of nitrate at floc surface	—	mg/L	M/L^3
SC6	Concentration of heterotrophs at floc surface	—	mg/L	M/L^3
SC7	Concentration of ammonia oxidizers at floc surface	—	mg/L	M/L^3
SC8	Concentration of nitrite oxidizers at floc surface	—	mg/L	M/L^3
Y_H	Yield coefficient for heterotrophs using oxygen as electron acceptor	0.408	Biomass COD/methanol COC	M/M
Y_A	Yield coefficient for ammonia oxidizers	0.265	Biomass COD/ammonium-nitrogen	M/M
Y_N	Yield coefficient for nitrite oxidizers	0.077	Biomass COD/nitrite-nitrogen	M/M
Y_{HNI}	Yield coefficient for heterotrophs using nitrite as electron acceptor	0.42	Biomass COD/methanol COD	M/M
Y_{HNA}	Yield coefficient for heterotrophs using nitrate as electron acceptor	0.35	Biomass COD/methanol COD	M/M
K1	Maximum rate of methanol use by heterotrophs using oxygen as electron acceptor for energy and biosynthesis	7.15	g methanol COD/g biomass COD · d	1/T
K2	Maximum rate of ammonium use by ammonia oxidizers for energy and biosynthesis	1.5	g ammonium-nitrogen/g biomass COD · d	1/T
K3	Maximum rate of nitrite use by nitrite oxidizers for energy and biosynthesis	6.41	g nitrite-nitrogen/g biomass COD · d	1/T
K4	Fractional loss of viable biomass per unit time for heterotrophs using oxygen as electron acceptor	0.1	g biomass COD/g biomass COD · d	1/T
K5	Maximum rate of methanol use by heterotrophs using nitrite as electron acceptor for energy and biosynthesis	7.79	g methanol COD/g biomass COD · d	1/T
K6	Fractional loss of viable biomass per unit time for heterotrophs using nitrite as electron acceptor	0.1	g biomass COD/g biomass COD · d	1/T
K7	Maximum rate of methanol use by heterotrophs using nitrate as electron acceptor for energy and biosynthesis	5.48	g methanol COD/g biomass COD · d	1/T
K8	Fractional loss of viable biomass per unit time for heterotrophs using nitrate as electron acceptor	0.1	g biomass COD/g biomass COD · d	1/T
KS1	Half-velocity constant for dissolved oxygen effect on heterotrophs	0.04	mg O_2/L	M/L^3
KS2	Half-velocity constant for dissolved oxygen effect on ammonia oxidizers	0.97	mg O_2/L	M/L^3
KS3	Half-velocity constant for dissolved oxygen effect on nitrite ozidizers	0.87	mg O_2/L	M/L^3
KS4	Half-velocity constant for methanol effect on heterotrophs	18	mg COD/L	M/L^3
KS5	Half-velocity constant for ammonium effect on ammonia oxidizers	0.1	mg ammonia-nitrogen/L	M/L^3
KS6	Half-velocity constant for nitrite effect on nitrite ozidizers	2.51	mg nitrite-nitrogen/L	M/L^3

be the concentrations at the floc surface), which varied from run to run, and the heterotroph concentration, which was used as the single parameter to calibrate the model. The diameter of the floc particle also varied from experiment to experiment. Table 5 summarizes the conditions for the six experimental runs, along with the heterotroph concentration determined through model calibration. The concentrations of ammonia- and nitrite-oxiders (model default values) are also shown for comparison.

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Table 3—Results of sensitivity analysis.*

	High sensitivity (Ratio $\geq$ 1)	Moderate sensitivity (0.1 $\leq$ Ratio $<$ 1)	Low sensitivity (Ratio $<$ 0.1)
Dissolved oxygen	D1, SC1	SC2, SC6, Y_H, K1	SC3, SC4, SC7, Y_A, K2, K3, K4, KS1, KS2, KS3, KS4, KS6
Methanol	SC2		SC1, SC6, K1, K5, D2, K7
Ammonium	SC3, D3		SC1, SC2, SC6, SC7, K1, K2
Nitrite	SC1, SC4	D3, SC2, SC3, SC6, SC7, SC8 K2, K3, K5, KS1, KS6	Y4, K6, KS3, KS2, KS4
Nitrate	D5, SC1	SC2, SC4, SC5, SC6, SC8, K3, K7, KS1, KS3, KS6	Y_{HNA}, K8, KS4
Heterotrophs	SC6	D6	SC1, SC2, Y_H
Nitrosomonas	D7, SC7		SC1, SC3, SC6, Y_{AO}, K2, K4, KS1, KS2
Nitrobacter	D8, SC8	SC1, SC6, K4	SC4, Y3, K3, KS1, KS3, KS6

* Parameters are defined in Table 2.

Figure 5 shows the results from the calibration run for sample A, including both the measured values and the calibrated model results. The results show that the depletion of dissolved oxygen from the bulk concentration of 0.74 mg/L occurred approximately 0.4 mm into the 3.0-mm diameter floc particle. This indicates that the dissolved oxygen concentration for approximately 40% of the floc volume was near zero, based on the assumption of a spherical floc particle. The results also show excellent comparison between the experimental data and the calibrated model. The R^2 value (or regression coefficient) for the model compared with the experimental data is 96.1%. Figure 6 shows the results for the second floc particle, 3.6 mm in diameter with a bulk dissolved oxygen concentration of 2.9 mg/L, examined from sample A. Again, the

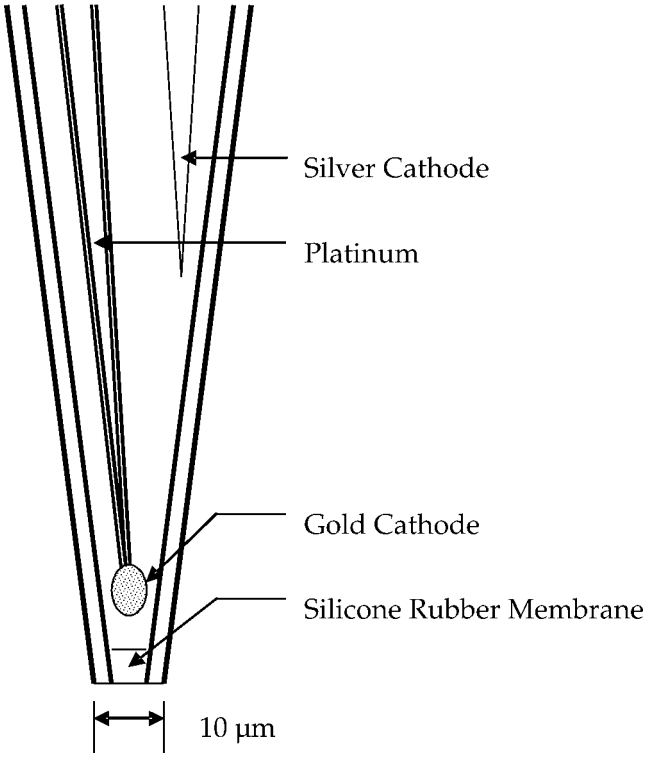


Figure 1—Unisense (Unisence ApS, Science Park Aarhus, Denmark) dissolved oxygen microprobe (guard cathode not shown).

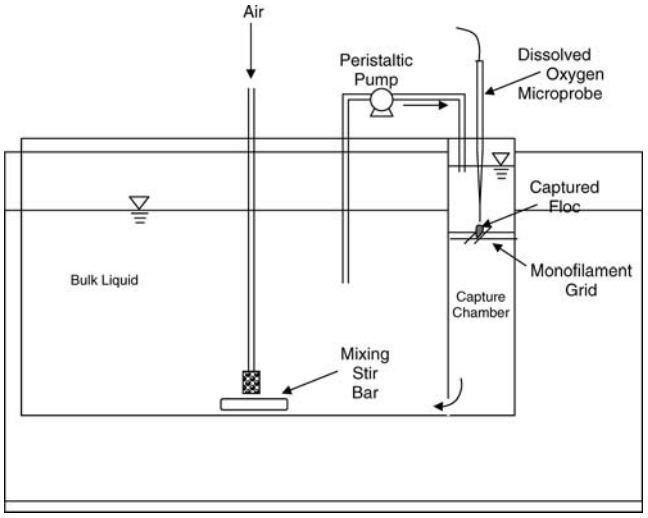


Figure 2—Schematic diagram of reactor setup used to capture floc and maintain bulk conditions (side view).

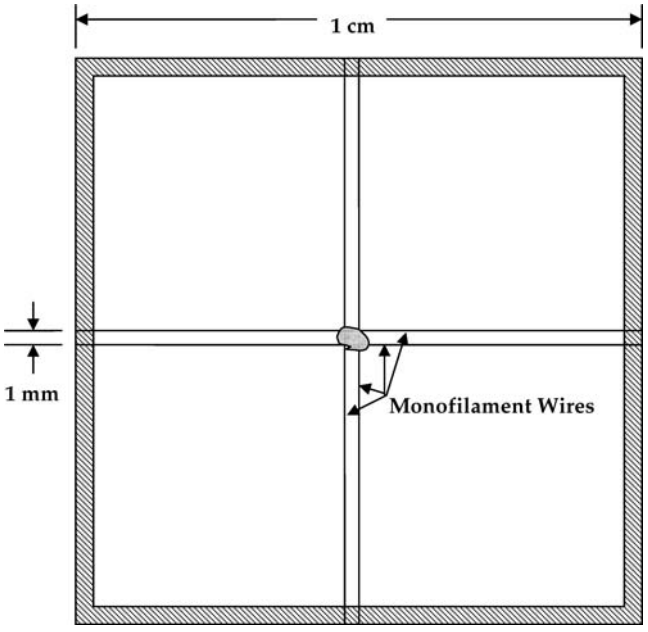


Figure 3—Wire grid used to capture activated sludge floc for dissolved oxygen profile measurement.

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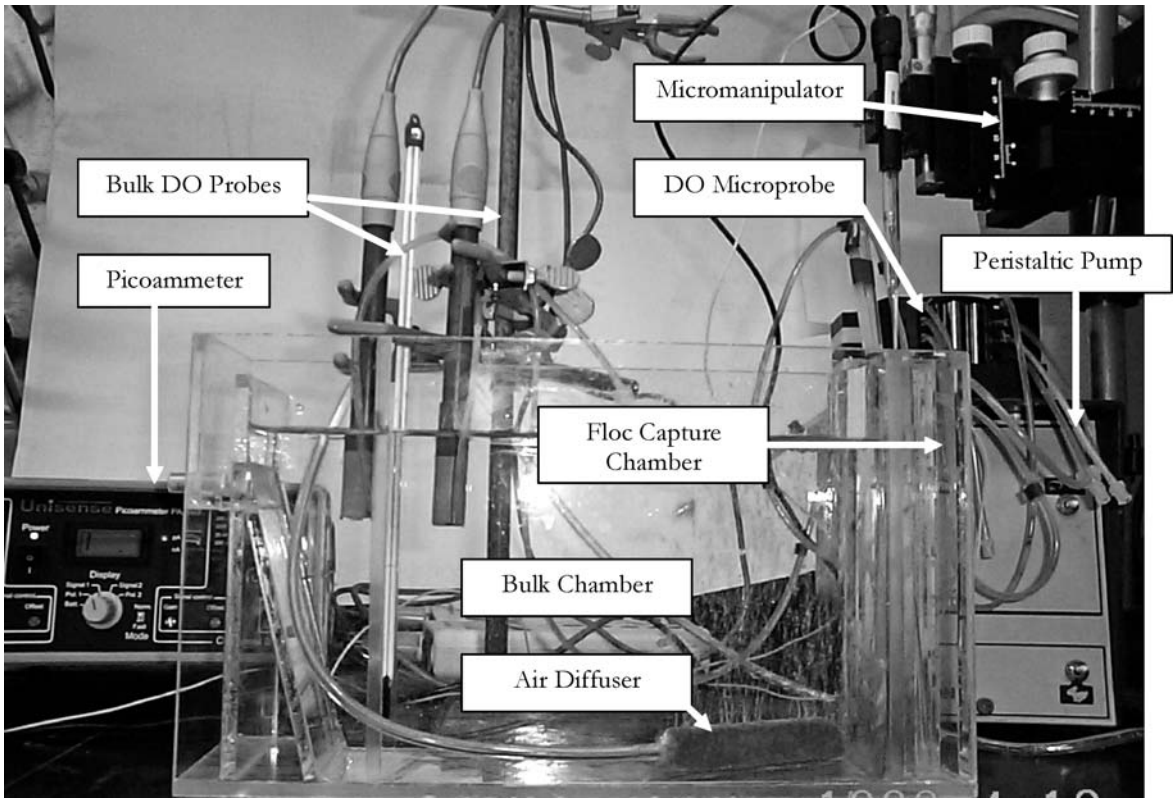


Figure 4—Reactor setup used for floc capture and intrafloc dissolved oxygen (DO) profile measurement.

dissolved oxygen concentration reached zero, this time at a depth of approximately 0.9 mm, which corresponds to a floc volume of approximately 12%, with near-zero dissolved oxygen based on the spherical floc particle assumption. The fit of the experimental data with the model calibrated to the other floc particle is also excellent, with an R^2 value of 98.2%.

Figure 7 shows the results for the first sample B floc particle, which has a 2.5-mm diameter. The dissolved oxygen concentration decreased from the bulk dissolved oxygen concentration of 2.9 mg/L to near zero, at a depth of approximately 0.3 mm, indicating that approximately 31% of the floc volume had a near-zero dissolved oxygen concentration based on the spherical floc particle assumption. As shown in Figure 7, the model could be fit to the model by adjusting the heterotroph concentration to 1500 mg/L (versus 200 mg/L for sample A) and that an excellent fit could be achieved. The R^2 for the calibrated model was 99.3%. Figure 8 shows a profile for the second floc particle (1-mm diameter) from sample B, which was examined at a bulk dissolved oxygen concentration of 5.8 mg/L. In this case, the measured dissolved oxygen profile exhibited a different shape than observed with the three previous profiles (Figures 5 to 7) and tended to level out at a depth of approximately 0.4 mm. The observed rate of dissolved oxygen depletion was intermediate between the model predictions based on the heterotroph concentrations determined for samples A and B. The R^2 for this floc particle compared with the sample B model was 88.6%, but exhibited an obvious lack of fit.

Figures 9 and 10 show the dissolved oxygen profiles for the sample C and D particles. The sample C particle (Figure 9) was 1.1 mm in diameter, and the bulk dissolved oxygen concentration was 3.5 mg/L, while the sample D particle was 4 mm in diameter, and the bulk dissolved oxygen concentration was 6.1 mg/L. The sample C floc particle showed a dissolved oxygen profile similar to that from validation run from sample B—declining at a rate intermediate between the predicted values based on the calibrated models from samples A and B and reaching a plateau value approximately 0.35 mm into the floc particle. In contrast, the larger sample D floc particle initially showed a decline in dissolved oxygen concentration that was well predicted by the model results using the sample A calibration, but diverged significantly at approximately 0.7 mm in depth, even though a near-zero dissolved oxygen concentration was reached at approximately 1.6 mm into the floc particle.

Table 4—Composition of synthetic solution used for floc dissolved oxygen profile measurements.

Component	Concentration (mg/L)
Methanol (as COD)	2000 to 3000
Ammonium chloride	280 to 300
Potassium phosphate, mono	2610
Potassium phosphate, di	2040
Sodium bicarbonate	2100
Magnesium sulfate	200
Calcium chloride	50
Ferric chloride	20
Ethylenediaminetetraacetic acid (EDTA)	30

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Table 5—Parameter values used for experimental runs.

Parameters	Run A		Run B		Run C	Run D
	Calibration	Validation	Calibrations	Validation		
Floc radius, R (mm)[a,b]	1.5	1.8	1.25	0.5	0.55	2.0
Bulk dissolved oxygen concentration (mg/L)[a,b]	0.74	2.9	2.9	5.8	3.5	6.1
Bulk methanol concentration (mg/L as COD)[a,b]	2061	2061	2.702	2.702	2310	2702
Bulk ammonia concentration (mg/L as N)[a,b]	301	301	301	280	282	280
Bulk nitrite concentration (mg/L as N)[a,b]	1	1	1	1	1	1
Bulk nitrate concentration (mg/L as N)[a,b]	1	1	1	1	1	1
Heterotroph concentration at floc surface (mg/L)	200[c]	200[c]	1500[c]	1500[c]	200[c] and 1500[d]	200[c] and 1500[d]
Ammonia-oxidizer concentration at floc surface (mg/L)[e]	27	27	27	27	27	27
Nitrite-oxidizer concentration at floc surface (mg/L)[e]	9	9	9	9	9	9
Sample date	2/10/2002	2/10/2002	2/12/2002	2/12/2002	2/14/2002	2/12/2002

[a] Measured experimentally.
[b] Concentration in the fluid recirculating past floc, assumed to be the same as at the floc surface.
[c] Adjusted for model calibration.
[d] Calibrated model value.
[e] Model default value.

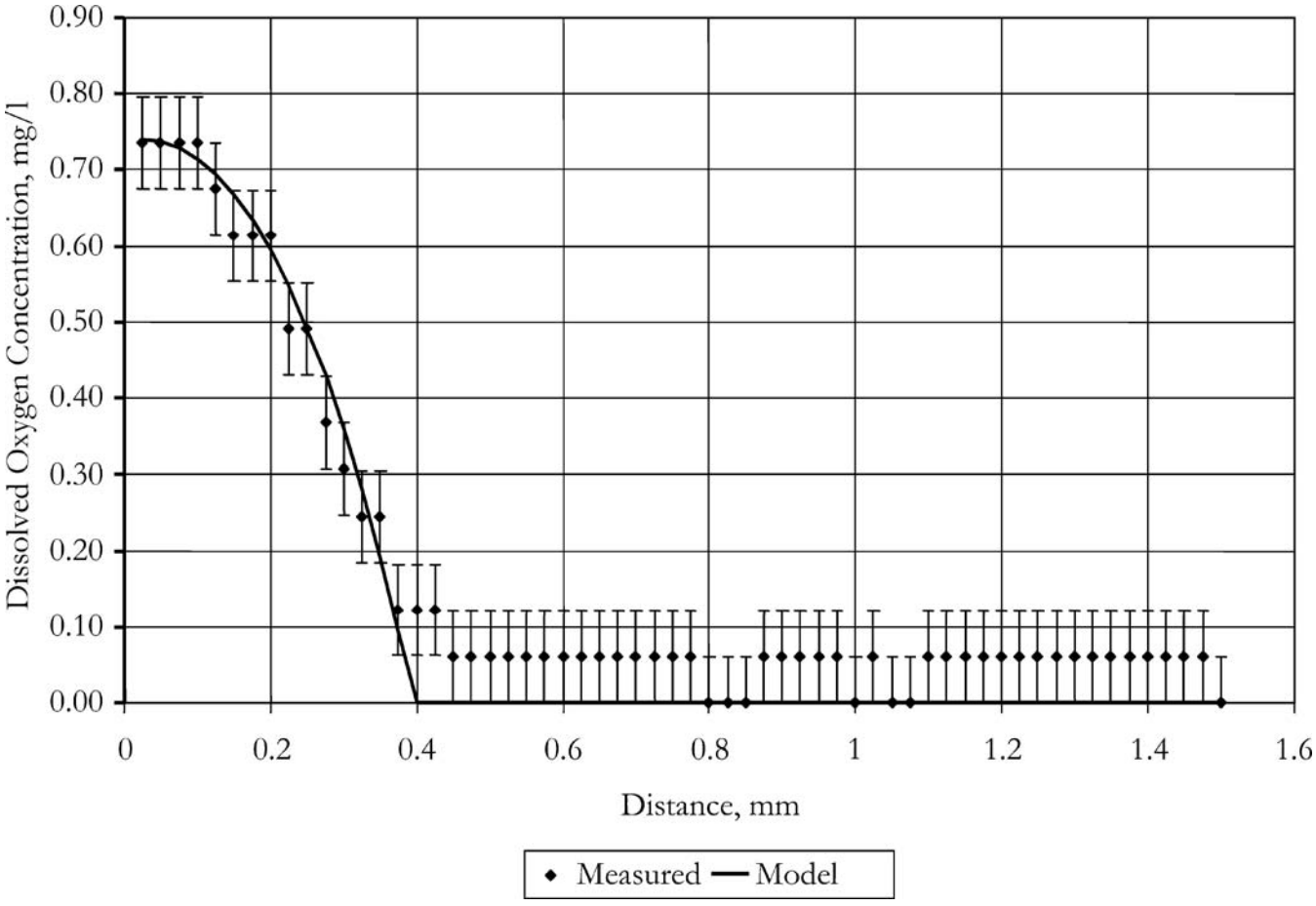


Figure 5—Dissolved oxygen profile for 3-mm floc particle from sample A and calibration to model.

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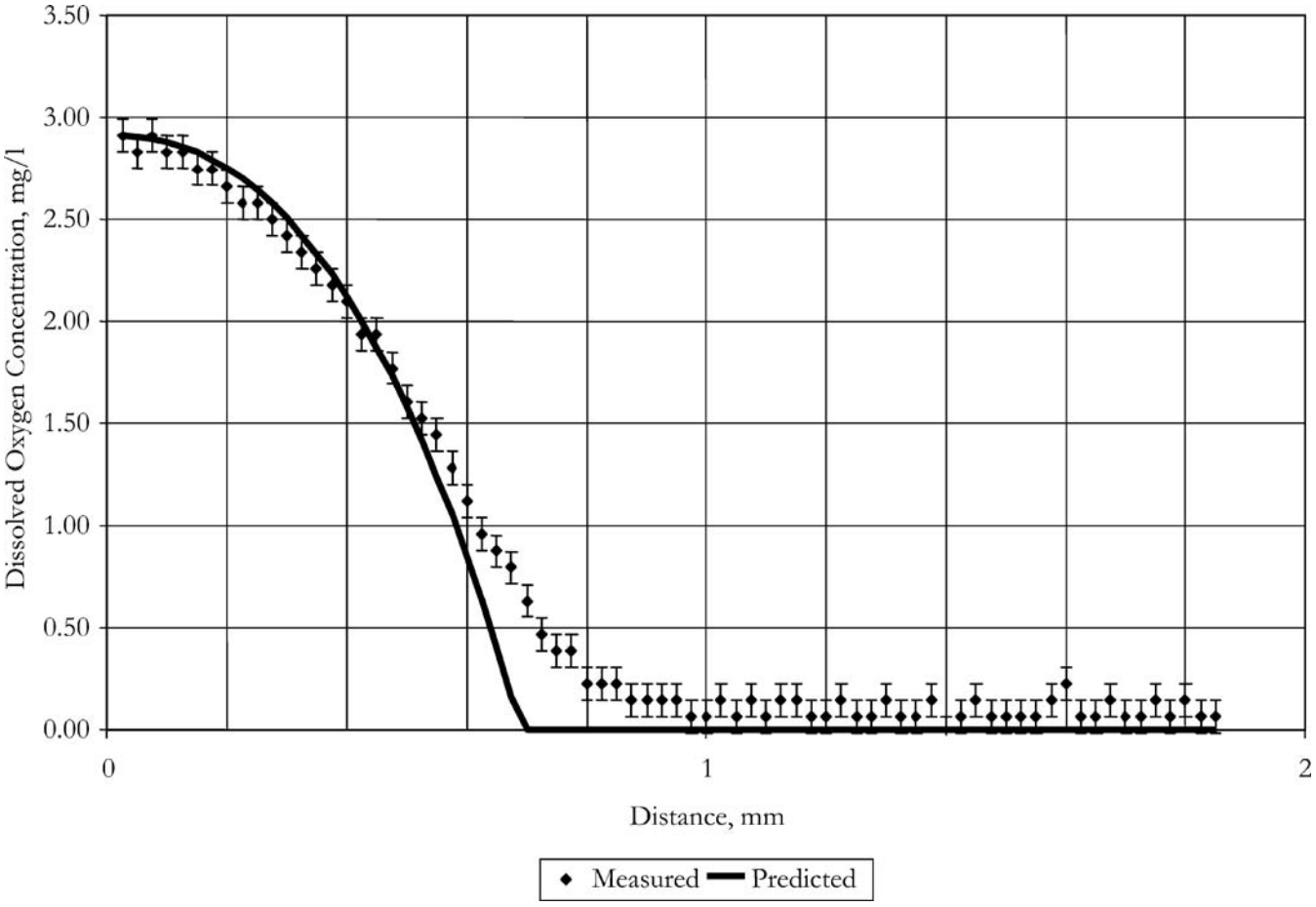


Figure 6—Dissolved oxygen profile for 3.6-mm floc particle from sample A used to validate sample A model.

Discussion

Similar dissolved oxygen profiles are observed for larger floc particles (2-mm diameter and larger), as illustrated in Figures 5 to 7 and 10, compared with the smaller flocs (approximately 1-mm diameter), as illustrated in Figures 8 and 9. For the larger flocs, the dissolved oxygen decreased regularly with depth and reached near-zero values. The penetration depth for dissolved oxygen depended on the external dissolved oxygen and the activity of the activated sludge floc (as represented in the concentration of heterotrophs in the calibrated model). The exception to this general trend is for the sample D floc particle, which exhibited this behavior for the outer portion of the activated sludge floc, but departed for the inner portion. It is uncertain why this divergence occurred, but it could have been a result of measurement difficulty, nonhomogeneity for the floc particle, or divergence from the assumption of a spherical floc particle. Li and Bishop (2004) observed similar dissolved oxygen profiles in floc particles with diameters greater than 1 mm from two wastewater treatment plants in Cincinnati, Ohio. In contrast, for the smaller floc particles (Figures 8 and 9), the dissolved oxygen decreased with depth into the floc particle, but tended to plateau at nonzero dissolved oxygen levels at the middle of the floc particle. The external dissolved oxygen was relatively high for both of these profiles (5.8 mg/L for Figure 8 and 3.5 mg/L for Figure 9), which may help to explain this behavior. Alternatively, these results may represent a difference in the structure and/or density of smaller versus larger floc particles. It is also possible that insertion of the dissolved oxygen microprobe disrupted the floc structure of the smaller particles (the probe was of fixed dimensions and, consequently, could disrupt smaller flocs to a greater extent than larger flocs), resulting in nonrepresentative dissolved oxygen profile measurements.

The mathematical model was able to characterize the dissolved oxygen profile through the larger floc particles quite well, by only adjusting the concentration of heterotrophs (Figures 5 to 7). For sample A, the model calibrated to the first floc particle (Figure 5) predicted the dissolved oxygen profile through the second floc particle (Figure 6) quite well. The model also characterized the dissolved oxygen profile through the first floc particle from sample B, although the heterotroph concentration had to be adjusted. This is reasonable, as the two samples were taken at different times from the full-scale wastewater treatment plant, and operating conditions could well have been quite different for the two samples. However, the profiles for the smaller floc particles (Figures 8 and 9) could not be characterized using the model. Efforts to adjust model parameters to allow the model to predict a leveling-off of the dissolved oxygen profile in the interior of the floc particle were unsuccessful (Steller, 2003).

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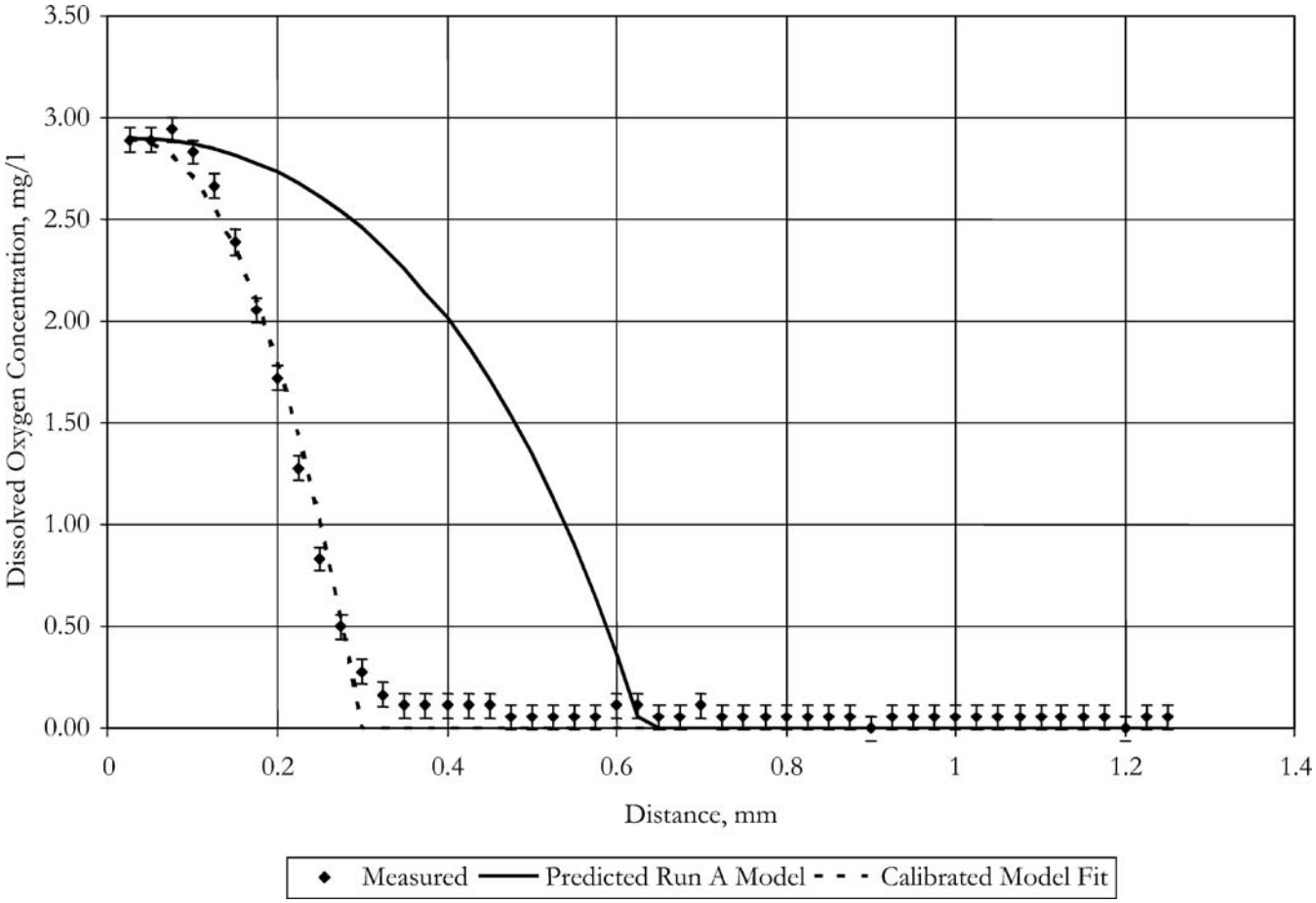


Figure 7—Dissolved oxygen profile for 2.5-mm floc particle from sample B and calibration to model. Predicted profile for sample A model shown for comparision.

The potential that different behaviors will be observed for floc particles of different sizes is significant, because the floc size in activated sludge systems is not uniform. Rather, a distribution of particle sizes exists (Jorand et al., 1995; Li and Ganczarczyk, 1991), with the specific distribution depending on a number of biological, chemical, and physical factors (Bouyer et al., 2001; Govoreanu et al., 2004; Li and Ganczarczyk, 1993; Wilen, 1999). For example, turbulence in the aeration basin can lead to shear that reduces particle size, while flocculation will increase particle sizes (Parker et al., 1971). A complete analysis of intrafloc effects on the behavior of an activated sludge process would require consideration of the distribution of floc particle sizes.

It is encouraging that model calibration required only adjustment of the heterotroph concentration. The sensitivity analysis (Table 3) indicated that the model was most sensitive to floc surface concentrations, followed by diffusivities for the diffusing species, and then kinetic parameters (maximum specific rates and half-velocity coefficients). For example, of the 14 parameters found to affect model response to the greatest extent (high sensitivity), 9 were floc surface concentrations, and 5 were diffusivities. Likewise, of the 29 parameters found to affect model response moderately (moderate sensitivity), 14 were surface concentrations, 2 were diffusivities, and 12 were kinetic parameters. When modeling an actual operating system using, for example, one of the IWA activated sludge models, many of these constituents and parameters, including the composition of the biomass, will be known. Thus, a diffusion model can be used to predict phenomena occurring inside the floc particles, assuming that values of the diffusivities are accurately known. In fact, the respiration rate predicted by the ASM for the operating conditions experienced can be used, with a dispersion model, to determine whether, for a particular operating bulk dissolved oxygen concentration, the activated sludge floc will remain fully aerobic or whether an oxygen-deficient inner portion will develop where denitrification can occur. Thus, one can envision a coupling of the ASM with an intrafloc diffusion model to characterize the intrafloc mechanism for simultaneous nitrification and denitrification.

The profiles presented in Figures 5 to 10 clearly illustrate that the concentrations of various substrates experienced by bacteria in activated sludge flocs can vary significantly from the bulk concentration. This requires a reinterpretation of the meaning of half-saturation coefficients for activated sludge systems. For example, it is possible that the observed half-saturation value for dissolved oxygen reflects the bulk dissolved oxygen needed to cause oxygen to fully penetrate the floc particle, rather than the actual effect of dissolved oxygen on the metabolism of the bacteria. Further, this dissolved oxygen will depend on a number of factors, including the respiration rate for the mixed liquor and floc particle size (and its distribution). Floc diffusion models can be used to estimate these

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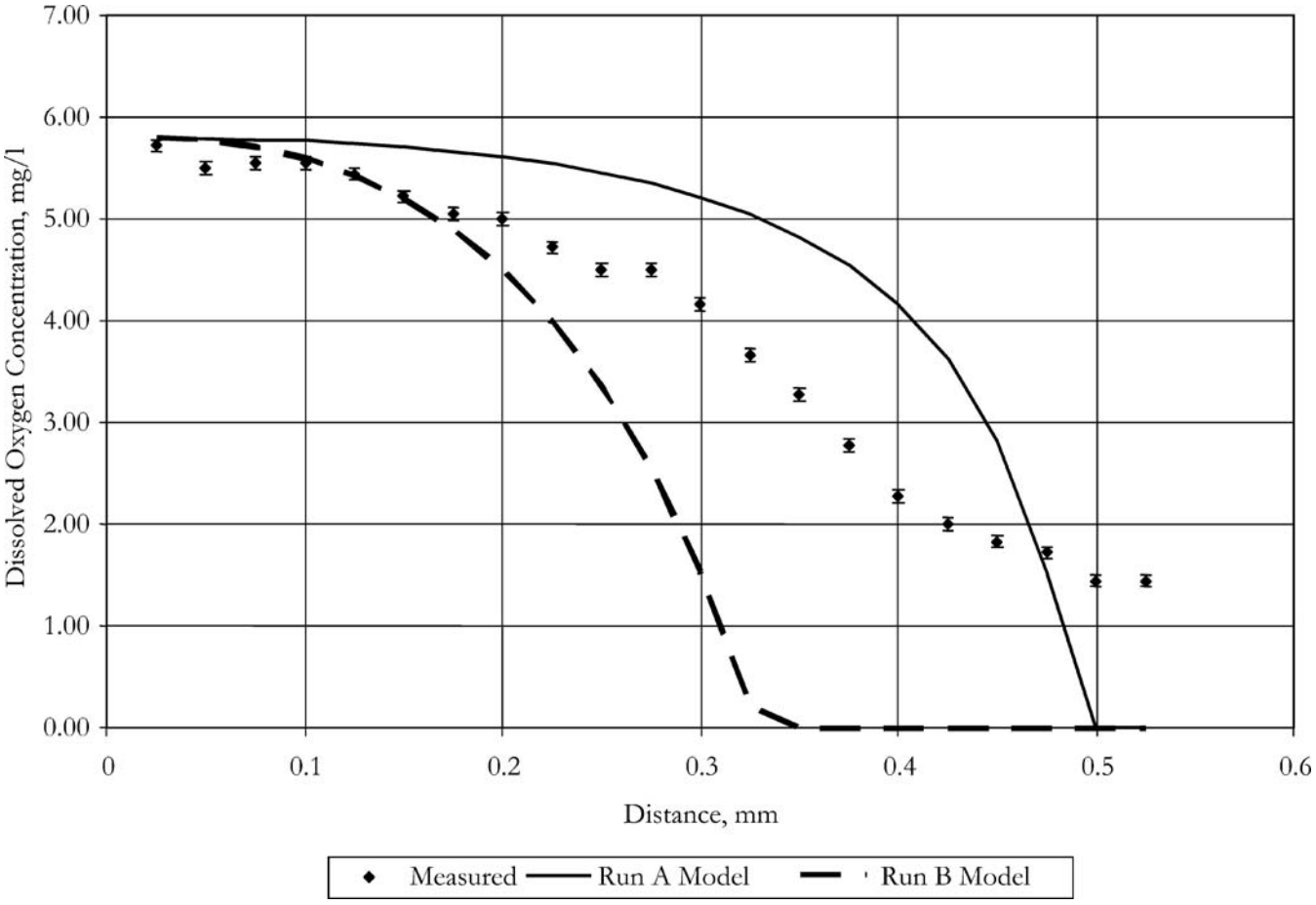


Figure 8—Dissolved oxygen profile for 1-mm floc particle from sample B and comparison with sample A and B model predictions.

effects. Moreover, because the respiration rate varies through many activated sludge systems (i.e., those possessing some degree of plug-free behavior), the half-saturation constant for various substrates can vary systematically through such a system.

Summary and Conclusions

In summary, an intrafloc diffusion model was developed to predict dissolved oxygen profiles through activated sludge flocs. This model was then compared with activated sludge floc dissolved oxygen profiles measured with a dissolved oxygen microprobe. For large floc particles (approximately 2-mm diameter or greater), a regular dissolved oxygen profile was observed that was well-predicted by the developed model. Model calibration was accomplished by adjusting a single model parameter—the concentration of heterotrophs. This concentration varied from one activated sludge sample to another, but was consistent from one floc particle to another for a specific activated sludge sample. The profiles for the larger floc particles also clearly demonstrated that dissolved-oxygen-deficient zones can form in activated sludge flocs. These are locations where denitrification can occur.

The dissolved oxygen profile through smaller floc particles also showed a decline in dissolved oxygen, but with a leveling-off, so that near-zero dissolved oxygen concentrations were not reached at the center of the floc. The mathematical model could not be adjusted to predict intrafloc dissolved oxygen profiles of this type.

Based on these results, the following can be concluded:

(1) The dissolved oxygen declines through activated sludge flocs of all sizes measured, so that the microorganisms in floc particles will experience dissolved oxygen concentrations lower than the bulk dissolved oxygen concentration. The decline in dissolved oxygen concentration in the floc particle varies, depending on both floc size and environmental conditions, so that the bulk dissolved oxygen concentration, by itself, is not an accurate indicator of the dissolved oxygen concentration that the microorganisms in an activated sludge floc will experience.
(2) The dissolved oxygen profile for the larger floc particles considered in this research declined and reached near-zero values at depths depending on floc properties and operating conditions. It is hypothesized that these near-zero zones within the floc can provide an environment where denitrification can occur.
(3) The dissolved oxygen decline within larger floc particles (2-mm diameter and greater) was accurately characterized using the mathematical model developed. In this model, dissolved oxygen, organic substrate (methanol), ammonia, nitrite, and nitrate are transported by diffusion through a spherical particle,

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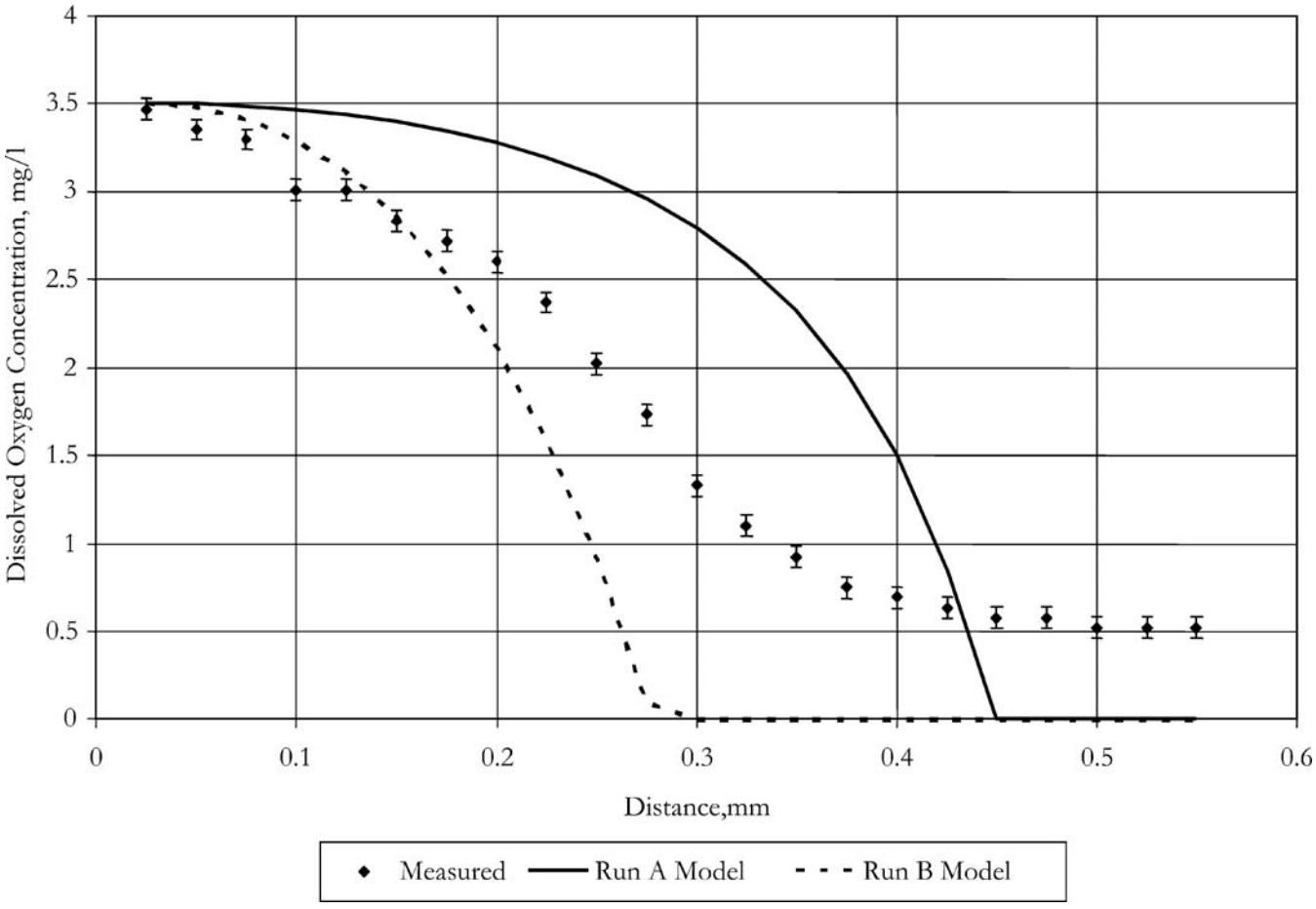


Figure 9—Dissolved oxygen profile for 1.1-mm floc particle from sample C and comparison with sample A and B model predictions.

and dissolved oxygen is consumed by oxidation of methanol by heterotrophs and ammonia and nitrite by autotrophs.

(4) The developed mathematical model could be calibrated to the dissolved oxygen profiles for the larger floc particles, by simply adjusting the concentration of heterotrophs. For a given activated sludge sample, the same model (same heterotroph concentration) was able to accurately characterize the dissolved oxygen profile through two separate activated sludge flocs. However, different heterotroph concentrations were required to characterize the dissolved oxygen profile through large floc particles through two sludge samples collected at different times, representing two different operating conditions for the full-scale activated sludge system.

(5) In contrast, for smaller floc particles (diameter of roughly 1 mm and smaller) the dissolved oxygen decreased, but did not reach near-zero values. Profiles of this shape could not be predicted by the developed mathematical model. This suggests that additional mechanisms are transporting dissolved oxygen into the activated sludge floc.

These results are significant relative to the occurrence of simultaneous nitrification and denitrification in activated sludge systems. The distribution of floc sizes in activated sludge systems must be considered in connection with the intrafloc denitrification mechanism. Factors that increase the average floc size, especially the proportion of large floc particles, will increase the potential that intrafloc denitrification can occur. For operating systems, the composition of biomass will be predicted by process mathematical models, such as the IWA activated sludge models. This composition, with an intrafloc diffusion model, can be used to calculate intrafloc dissolved oxygen profiles, to determine the bulk dissolved oxygen concentration required at various points within the process, to allow dissolved oxygen to decline to near-zero levels within the floc, thereby allowing intrafloc denitrification to occur. These results also suggest that apparent half-saturation constants for various substrates reflect both the inherent effect of substrate concentration on the activity of the microorganisms and the effects of diffusion of that substrate through the activated sludge floc particle. Consequently, the apparent half-saturation constants may vary through the activated sludge process, depending on process operating conditions and the resulting diffusion of that substrate into the activated sludge floc.

Credits

The authors acknowledge the support of the U.S. Department of Education (Washington, D.C.) for providing funding for Holley Kaempfer Steller during her doctoral studies through the Graduate Assistance in Areas of National Needs program (Grant No.

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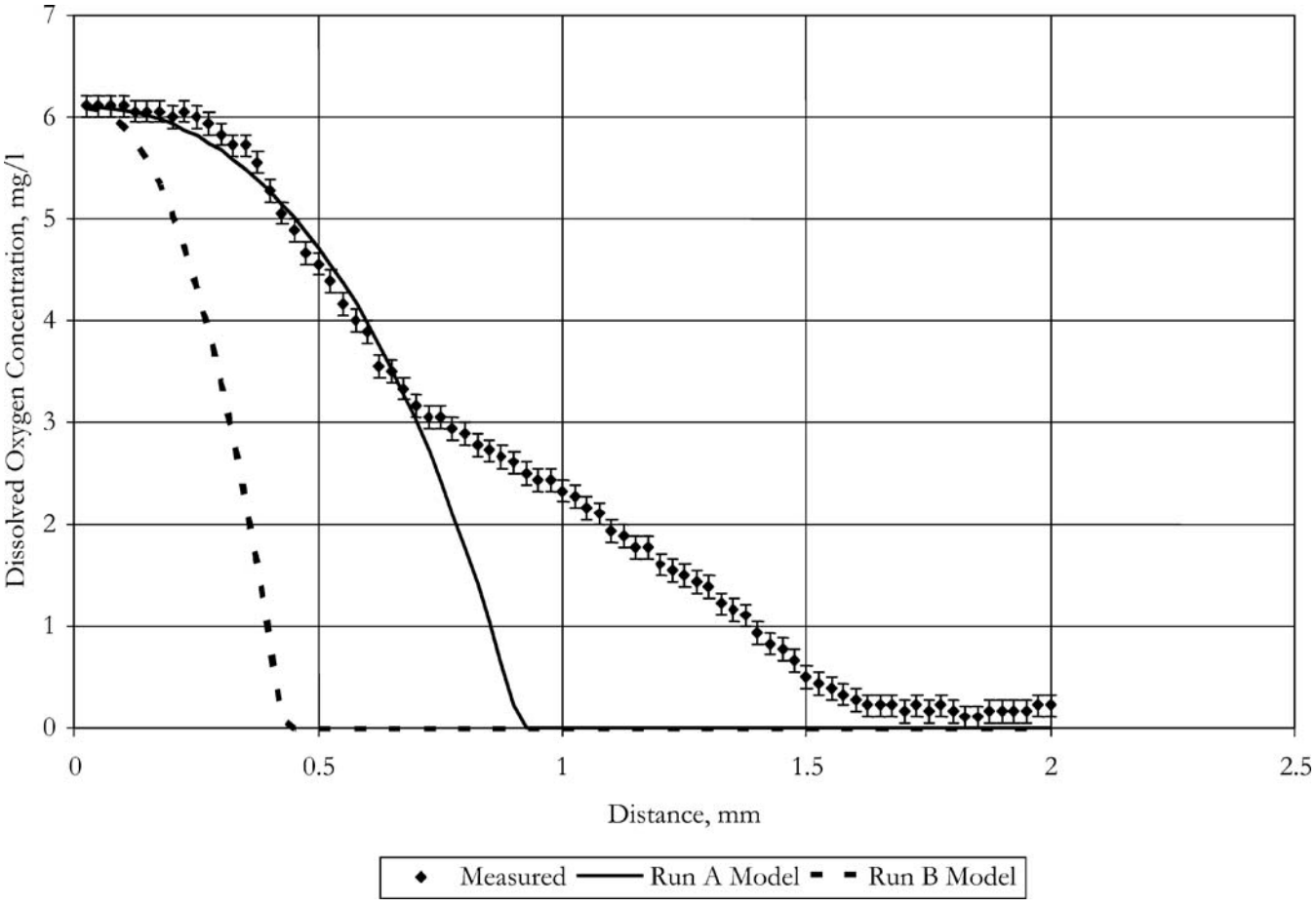


Figure 10—Dissolved oxygen profile for 4-mm floc particle from sample D and comparison with sample A and B model predictions.

P200A980418). The authors also thank Jeff Bradshaw (University of Missouri-Rolla) for assistance in electrical engineering and software interface aspects of the microprobe system.

Submitted for publication February 14, 2005; revised manuscript submitted March 19, 2006; accepted for publication March 27, 2006.

The deadline to submit Discussions of this paper is July 15, 2007.

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